

1. Show that $\Delta S = q_{\text{rev}}/T$.
2. Derive an expression for the entropy change accompanying the isothermal expansion of an ideal gas.
3. Show that the entropy of a system and surroundings taken together remains constant in a reversible process while the same increases in an irreversible process.
4. Comment on the statement. "Entropy of the universe is always increasing."
5. Derive expressions for the entropy change accompanying variation of (i) temperature and volume (ii) temperature and pressure.
6. Derive an expression for the entropy of a mixture of ideal gases.
7. Define the terms Gibbs free energy and Helmholtz free energy. How is each of these terms related to maximum work that can be done by a system during a given change? Discuss the variation of ΔG with variation in (i) temperature and pressure (ii) pressure and volume.
8. Derive Gibbs-Helmholtz equation for a process at constant pressure and at constant volume.

Example 11. One mole of an ideal gas expands reversibly from a volume of 10 dm^3 and temperature 298 K to a volume of 20 dm^3 and temperature 250 K . Assuming that $C_V = 3/2 R$, calculate the entropy change for the process.

Solution : From Eq. 42, $\Delta S = C_V \ln(T_2/T_1) + R \ln(V_2/V_1)$

$$= 1.5 (8.314 \text{ J K}^{-1} \text{ mol}^{-1}) \ln(250 \text{ K}/298 \text{ K}) + (8.314 \text{ J K}^{-1} \text{ mol}^{-1}) \ln(20 \text{ dm}^3/10 \text{ dm}^3)$$

$$= -2.19 \text{ J K}^{-1} \text{ mol}^{-1} + 5.76 \text{ J K}^{-1} \text{ mol}^{-1} = 3.57 \text{ J K}^{-1} \text{ mol}^{-1}$$

Example 12. If the gas in the above example expands reversibly and adiabatically, what would be the value of ΔS for the process?

Solution : Since for a reversible adiabatic process, $q_{\text{rev}} = 0$, hence $\Delta S = 0$.

Example 13. One mole of an ideal gas expands reversibly from a temperature of 25°C and pressure of 1 atm to a temperature of 0°C and pressure of 500 mm Hg . Calculate ΔS for the process. Assume that $C_V = 3/2 R$.

Solution : From Eq. 46, $\Delta S = C_P \ln(T_2/T_1) + R \ln(P_1/P_2)$

For an ideal gas, $C_P - C_V = R$

$$\therefore C_P = C_V + R = 3/2 R + R = 2.5 R = 2.5 \times 8.314 \text{ J K}^{-1} \text{ mol}^{-1}$$

$$\therefore \Delta S = (2.5 \times 8.314 \text{ J K}^{-1} \text{ mol}^{-1}) \ln(273 \text{ K}/298 \text{ K}) + (8.314 \text{ J K}^{-1} \text{ mol}^{-1}) \ln 760 \text{ mm Hg}/500 \text{ mm Hg}$$

$$= -1.82 \text{ J K}^{-1} \text{ mol}^{-1} + 3.48 \text{ J K}^{-1} \text{ mol}^{-1} = 1.66 \text{ J K}^{-1}$$

Example 3. 5 moles of an ideal gas expand reversibly from a volume of 8 dm^3 to 80 dm^3 at a temperature of 27°C . Calculate the change in entropy.

Solution : From Eq. 31, $\Delta S = nR \ln(V_2/V_1)$

$$= (5 \text{ mol}) (8.314 \text{ J K}^{-1} \text{ mol}^{-1}) \ln(80 \text{ dm}^3/8 \text{ dm}^3) = 95.73 \text{ J K}^{-1}$$

Example 17. Calculate the free energy change (ΔG) which occurs when 1 mole of an ideal gas expands reversibly and isothermally at 37°C from an initial volume of 55 dm^3 to 1000 dm^3 .

Solution : Substituting the given data in Eq. 90, we have

$$\Delta G = (8.314 \text{ J K}^{-1} \text{ mol}^{-1}) (310 \text{ K}) \ln(55 \text{ dm}^3/1000 \text{ dm}^3) = -7476.8 \text{ J mol}^{-1}$$

Example 9. Calculate the change in entropy accompanying the heating of one mole of helium gas, assumed ideal, from a temperature of 298 K to a temperature of 1000 K at constant pressure. Assume that $C_V = \frac{3}{2}R$.

Solution : Since for an ideal gas, $C_p - C_V = R$, hence, $C_p = C_V + R = \frac{3}{2}R + R = 2.5R$

When pressure is constant, Eq. 46 takes the form

$$\Delta S = C_p \ln (T_2/T_1) = (2.5 \times 8.314 \text{ J K}^{-1} \text{ mol}^{-1}) \ln (1000 \text{ K}/298 \text{ K}) = 25.17 \text{ J K}^{-1} \text{ mol}^{-1}$$

Example 10. One mole of an ideal gas is heated from 100 K to 300 K. Calculate ΔS if (a) the volume is kept constant (b) the pressure is kept constant Assume that $C_V = 1.5R$.

Solution : When volume is kept constant, Eq. 42 reduces to

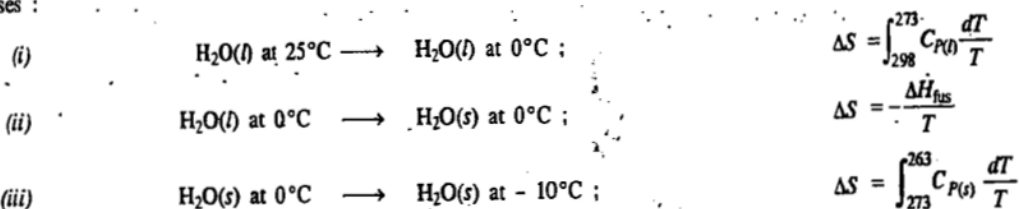
$$\Delta S = C_V \ln (T_2/T_1) = (1.5 \times 8.314 \text{ J K}^{-1} \text{ mol}^{-1}) \ln (300 \text{ K}/100 \text{ K}) = 13.17 \text{ J K}^{-1}$$

When pressure is kept constant, Eq. 46 takes the form

$$\begin{aligned} \Delta S &= C_p \ln (T_2/T_1) = (C_V + R) \ln (T_2/T_1) = 2.5 R \ln (T_2/T_1) \\ &= (2.5 \times 8.314 \text{ J K}^{-1} \text{ mol}^{-1}) \ln (300 \text{ K}/100 \text{ K}) = 22.8 \text{ J K}^{-1} \text{ mol}^{-1} \end{aligned}$$

Example 6. Calculate the entropy change accompanying the freezing of one mole of water at 25°C to ice at -10°C; given that the heat of fusion of ice at its fusion point (0°C) is 6.00 kJ mol⁻¹, the heat capacity of ice is 36.82 J K⁻¹ mol⁻¹ and heat capacity of liquid water is 75.31 J K⁻¹ mol⁻¹.

Solution : The irreversible freezing of water at -10°C can be split into the following three reversible processes :



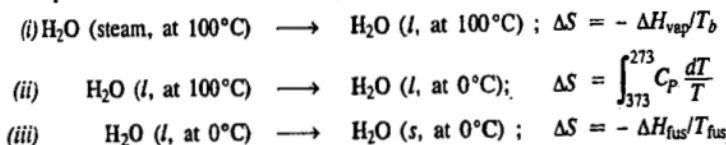
The total entropy change is the sum of the entropies involved in the three steps indicated above. Thus,

$$\begin{aligned} \Delta S_{\text{total}} &= \int_{298}^{273} (75.31 \text{ J K}^{-1} \text{ mol}^{-1}) \frac{dT}{T} - \frac{6000 \text{ J mol}^{-1}}{273 \text{ K}} + \int_{273}^{263} 36.82 \text{ J K}^{-1} \text{ mol}^{-1} \frac{dT}{T} \\ &= (75.31 \text{ J K}^{-1} \text{ mol}^{-1}) \ln \frac{T_2}{T_1} - \frac{6000 \text{ J mol}^{-1}}{273 \text{ K}} + (36.82 \text{ J K}^{-1} \text{ mol}^{-1}) \ln \frac{T_2}{T_1} \\ &= (75.31 \text{ J K}^{-1} \text{ mol}^{-1}) \ln \frac{273 \text{ K}}{298 \text{ K}} - 21.98 \text{ J K}^{-1} \text{ mol}^{-1} + (36.82 \text{ J K}^{-1} \text{ mol}^{-1}) \ln \frac{263 \text{ K}}{273 \text{ K}} \\ &= -29.96 \text{ J K}^{-1} \text{ mol}^{-1} \end{aligned}$$

Example 7. Calculate ΔS when one mole of steam at 100°C is converted into ice at 0°C. Average specific heat of water = $4.184 \times 10^3 \text{ J K}^{-1} \text{ kg}^{-1}$; heat of vaporisation at boiling point = $2490.6 \times 10^3 \text{ J kg}^{-1}$; heat of fusion at freezing point = $333.555 \times 10^3 \text{ J kg}^{-1}$.

$$\begin{aligned} \text{Solution :} \quad C_p &= (4.184 \times 10^3 \text{ J K}^{-1} \text{ kg}^{-1}) (18.0 \times 10^{-3} \text{ kg mol}^{-1}) = 75.31 \text{ J K}^{-1} \text{ mol}^{-1} \\ \Delta H_{\text{vap}} &= (2490.6 \times 10^3 \text{ J kg}^{-1}) (18.0 \times 10^{-3} \text{ kg mol}^{-1}) = 44831.56 \text{ J mol}^{-1} \\ \Delta H_{\text{fus}} &= (333.555 \times 10^3 \text{ J kg}^{-1}) (18.0 \times 10^{-3} \text{ kg mol}^{-1}) = 6004.00 \text{ J mol}^{-1} \end{aligned}$$

The process consists of the following reversible steps :



The total entropy is the sum of the three contributions. Thus,

$$\begin{aligned} \Delta S_{\text{total}} &= -\Delta H_{\text{vap}}/T_b + C_p \ln 273 \text{ K}/373 \text{ K} - \Delta H_{\text{fus}}/T_{\text{fus}} \\ &= -\frac{44831.56 \text{ J mol}^{-1}}{373 \text{ K}} + (75.31 \text{ J K}^{-1} \text{ mol}^{-1}) \ln \frac{273 \text{ K}}{373 \text{ K}} - \frac{6004.00 \text{ J mol}^{-1}}{273 \text{ K}} \\ &= -165.73 \text{ J K}^{-1} \text{ mol}^{-1} \end{aligned}$$

Example 29. The equilibrium constant K_p for the reaction $\text{H}_2(\text{g}) + \text{S}(\text{g}) \rightleftharpoons \text{H}_2\text{S}(\text{g})$ is 20.2 atm^{-1} at 945°C and 9.1 atm^{-1} at 1065°C . Calculate ΔH° .

Solution : According to the integrated van't Hoff equation

$$\ln \left(\frac{K_{p,2}}{K_{p,1}} \right) = \frac{\Delta H^\circ}{R} \left(\frac{T_2 - T_1}{T_1 T_2} \right)$$

or
$$\Delta H^\circ = R \ln \left(\frac{K_{p,2}}{K_{p,1}} \right) \left(\frac{T_1 T_2}{T_2 - T_1} \right) = (8.314 \text{ J K}^{-1} \text{ mol}^{-1}) \ln (9.21/20.2) \left[\frac{(1218 \text{ K})(1338 \text{ K})}{1338 \text{ K} - 1218 \text{ K}} \right]$$

$$= -88126.3 \text{ J mol}^{-1} = -88.126 \text{ kJ mol}^{-1}$$

Example 30. Integrate the van't Hoff equation, $d \ln K_p/dT = \Delta H^\circ/RT^2$ and determine the constant of integration.

Solution : $d \ln K_p/dT = \Delta H^\circ/RT^2$ or $d \ln K_p = (\Delta H^\circ/RT^2) dT$

Assuming that ΔH° remains constant over the temperature range, we have

$$\int (d \ln K_p) = \frac{\Delta H^\circ}{R} \int \frac{dT}{T^2} \quad \text{or} \quad \ln K_p = -(\Delta H^\circ/RT) + I$$

where I is the constant of integration.

Since $\Delta G^\circ = -RT \ln K_p$, we have

$$-\Delta G^\circ/RT = \ln K_p = -(\Delta H^\circ/RT) + I$$

or

$$\Delta G^\circ = \Delta H^\circ - IRT$$

Also,

$$\Delta G^\circ = \Delta H^\circ - T\Delta S^\circ$$

From Eqs. (i) and (ii), $I = \Delta S^\circ/R$

Example 31. The equilibrium constant of a reaction doubles on raising the temperature from 25°C to 35°C . Calculate ΔH° for the reaction.

Solution : According to the integrated van't Hoff equation, $\ln \left(\frac{K_{p,2}}{K_{p,1}} \right) = \frac{\Delta H^\circ}{R} \left(\frac{T_2 - T_1}{T_1 T_2} \right)$

Here

$$T_1 = 25^\circ\text{C} = 298 \text{ K}; T_2 = 308 \text{ K}$$

$$K_{p,2}/K_{p,1} = 2; R = 8.314 \text{ J K}^{-1} \text{ mol}^{-1}$$

$$\Delta H^\circ = R [T_1 T_2 / (T_2 - T_1)] \ln 2 = 52893.5 \text{ J} = 52.89 \text{ kJ mol}^{-1}$$

Example 32. For the reaction $\text{CO}(\text{g}) + \text{SO}_3(\text{g}) \rightleftharpoons \text{CO}_2(\text{g}) + \text{SO}_2(\text{g})$, $\Delta G^\circ = 187.1 \text{ kJ mol}^{-1}$ and $\Delta H^\circ = 184.7 \text{ kJ mol}^{-1}$ at 25°C . Calculate (a) ΔG° at 398 K and (b) K_p at 398 . Assume that ΔH° remains constant over the temperature interval.

Solution : (a) $\left[\frac{\partial(\Delta G^\circ/T)}{\partial T} \right]_P = -\frac{\Delta H^\circ}{T^2}$

(Eq. 63)

$$d(\Delta G^\circ/T) = -(\Delta H^\circ/T^2) dT$$

Integrating, we get, $\int_{298}^{398} d\left(\frac{\Delta G^\circ}{T}\right) = -\int_{298}^{398} \frac{\Delta H^\circ}{T^2} dT$

$$\left[\frac{\Delta G^\circ}{T} \right]_{298}^{398} = -\Delta H^\circ \left[-\frac{1}{T} \right]_{298}^{398} = \Delta H^\circ \left[\frac{1}{T} \right]_{298}^{398}$$

$$\frac{\Delta G_{398}^\circ}{398 \text{ K}} - \frac{\Delta G_{298}^\circ}{298 \text{ K}} = \Delta H^\circ \left[\frac{1}{398 \text{ K}} - \frac{1}{298 \text{ K}} \right]$$

Substituting the values of ΔG° and ΔH° at 25°C , we have

$$\frac{\Delta G^\circ}{398 \text{ K}} = \frac{187.1 \text{ kJ mol}^{-1}}{298 \text{ K}} - (184.7 \text{ kJ mol}^{-1}) \left(\frac{1}{398 \text{ K}} - \frac{1}{298 \text{ K}} \right)$$

$$= -0.472 \text{ kJ K}^{-1} \text{ mol}^{-1}$$

$$\Delta G_{398}^\circ = (-0.472 \text{ kJ K}^{-1} \text{ mol}^{-1})(398 \text{ K}) = -187.8 \text{ kJ mol}^{-1}$$

(b) Since

$$\Delta G^\circ = -RT \ln K_p$$

$$K_p = \exp \left(\frac{-\Delta G^\circ}{RT} \right) = 4.46 \times 10^{24}$$